PEACE ODIASE

MD/PHD CANDIDATE
PITT–CMU MSTP

Pittsburgh, PA
peo22@pitt.edu
peaceodiase.com
ORCID 0009-0007-5929-8795
Google Scholar

EDUCATION

- University of Pittsburgh & Carnegie Mellon University
  Present

MD/PhD Candidate, Medical Scientist Training Program

- Fisk University
  2012 – 2016

Bachelor of Arts in Biology

- Valedictorian, GPA 4.0
- Erastus Milo Cravath Presidential Scholar
- School Editorial Committee (President) | Every Nation Campus Ministries (President) | Academic Tutor (Math, Chemistry, Organic Chemistry)

RESEARCH EXPERIENCE

- Carnegie Mellon University
  June 2025 – Present

Goal Selection Under Context Uncertainty in Medical AI

SMASH Lab, Human-Computer Interaction Institute · PI: Mayank Goel

- Analyzed physician responses to patient questions on Reddit and identified recurring response archetypes: triage, clinical questioning, recommendations, and social support.
- Running user studies of a goal-aware system that infers what a response should accomplish for a patient and conditions generation on that goal.

- Carnegie Mellon University
  June – July 2026

Neural Expectations in Interactive Systems

Augmented Perception Lab, Human-Computer Interaction Institute · with D. Lindlbauer, A. Ion, S. Wood, and BMW Group

- Built a simulated driving environment in which the system covertly steered the wheel to elicit surprise.
- Recorded participant responses with 31-channel EEG, facial-affect tracking, and pupillometry; manuscript under review.

- Carnegie Mellon University
  June – July 2025

Evidotes: Evidence and Anecdotes for Peer Health Posts

SMASH Lab, Human-Computer Interaction Institute · PI: Mayank Goel

- Co-authored an LLM-powered browser extension that pairs Reddit health posts with PubMed evidence and peer anecdotes through three lenses: Dive Deeper, Focus on Positivity, and Big Picture.
- Published at CHI 2026; received an Honorable Mention Award (top 5%).

- Meharry Medical College
  August 2020 – 2025

Clinical Coordinator and Research Lead – Microbiology + Neuroscience

- Studied how cholesterol depletion affects dopamine transporter function using super-resolution (STORM) microscopy.
- Ran population studies of cholesterol-lowering medication and methamphetamine relapse.
- Co-authored work on the methamphetamine-induced interactome and SILAC quantification of proteins in dopaminergic cells.
- Studied nasal microbiome profiles and COVID-19 severity in immunocompromised patients with HIV; contributed to glioblastoma outcomes research.

- Fisk University
  May 2014 – December 2018

Research Volunteer – School of Graduate Science

- Investigated how *hsp12.6* regulates dopamine signaling via an *fkh-8*-mediated pathway in *C. elegans*, and how *dat-1* missense alleles affect transporter function and localization.
- Functionally characterized variants from the Million Mutation Project and confirmed DAT-1 loss-of-function behaviors through forward genetic screening.

- Vanderbilt University
  June – August 2014

Research Intern – Vanderbilt Summer Undergraduate Research Experience

Investigated the role of the novel gene *swip-10* in regulating dopamine signaling through a glutamatergic pathway in *C. elegans*, resulting in a presented poster and co-authorship in the *Journal of Neuroscience*.

- Meharry Medical College
  May 2013 – January 2014

Research Intern – Prostate Cancer Training Program

Investigated how androgen-regulated oncomirs miR-27a/b affect PPARγ levels in human prostate cancer cells, and designed a community-based survey on health disparities in newborn weight and prostate cancer screening.

- Fisk University
  September 2012 – April 2013

Research Volunteer – Polymer Chemistry

Synthesized polyxylitol sebacate (PXS) for skin grafting and tissue engineering, and used it as a plasticizer in chitosan-based elastomers to develop biodegradable plastics.

INDUSTRY & ENTREPRENEURSHIP

- Kythco
  2026 – Present

Co-Founder (with Shidhartho Roy)

Building an at-home care app for people with Parkinson's disease, with brief check-ins tied to medication timing and compared with each person's own baseline.

- Momentive Software
  (formerly Community Brands)
  May 2022 – 2025

Analytics Manager – B2B

Achieved 41% year-over-year growth in qualified inquiries and 20% in pipeline for global medical organizations by optimizing customer journeys, improving data visibility across channels, and refining market research.

- Osaze
  November 2022 – 2025

Co-Founder & Chief Data Analyst – B2B2C

Led Osaze from concept to launch, driving year-over-year revenue growth through sales, marketing, and operations while establishing key partnerships and building the founding team.

- Acadia Healthcare
  July 2021 – May 2022

Corporate Analyst – B2C

Drove a 110% year-over-year increase in access to eating disorder facilities by guiding decision-making for 77 mental healthcare companies, optimizing patient targeting, and improving satisfaction and conversions.

- Fisk University
  July 2019 – July 2020

Supplemental Instruction Supervisor

Coordinated the Supplemental Instruction program, including recruiting, training, and developing program instructors.

- Allscripts EHR (now Veradigm)
  June – August 2015

Instructional Design Intern

Responsible for end-user testing and front-end development of a patient enrollment portal.

- Freelance
  Seasonal

Freelance Web Developer

Design and build websites for individuals and organizations, including dnpt.org and peaceodiase.com.

PUBLIC HEALTH & ADVOCACY

- Catering to Africans in Need
  April 2013 – Present

Founding Member and Director of Global Partnerships

Helped establish a clinic in Jeida, Nigeria, train community health workers, and create the first free educational resource center in Alimosho; programs have reached more than 10,000 children and over 1,000 students through the CAIN Career Enrichment Program.

- United to Beat Malaria – United Nations Foundation
  September 2023 – 2025

Advocacy Champion – Tennessee

Discussed advocacy strategies for federal malaria funding, met with U.S. lawmakers and congressional offices, and mobilized grassroots support through community organizations.

- The M. Kukoyi Foundation
  June 2020 – February 2024

Director of Global Operations

Launched programs that trained more than 400 community members, improved traumatic brain injury survival from 50% to 94% across 14 hospitals, and supplied low-resource centers.

- Distinguished Nigerian Physicians of Tomorrow
  September 2020 – June 2024

President; Research and Outreach Chair

Organized the annual scientific conference, bimonthly physician mentorship meetings, fundraising, research efforts, and mission trips to Nigeria.

- Salt Wagon Clinic
  November 2020 – December 2023

Director of Media Relations, Clinical Shadowing

Coordinated a flu vaccination event with the Metropolitan Public Health Department, addressed COVID-19 disparities in minority groups, and managed social media and a scheduling portal.

PUBLICATIONS

UNDER REVIEW & IN PROGRESS

1. S. Roy, **P. Odiase**, A. Kulshrestha, K. Tsang, S. Heijboer, E. Iliffe-Moon, B. Mok, S. Pandya, S. Wood, A. Ion, D. Lindlbauer. Towards Understanding and Modeling Neural Expectations in Interactive Systems. *Under review*, 2026.
2. What Should the Answer Accomplish? Goal Selection Under Context Uncertainty in Medical AI. *In progress (user studies underway)*, SMASH Lab, Carnegie Mellon University.

PEER-REVIEWED · Google Scholar Profile

1. S. Bali, R. Arakawa, **P. Odiase**, T. Wu, M. Goel. Evidotes: Integrating Scientific Evidence and Anecdotes to Support Uncertainties Triggered by Peer Health Posts. *Proc. 2026 CHI Conference on Human Factors in Computing Systems*, 2026. **Honorable Mention Award (top 5%).**
2. S. Srinivasan, M. M. Eraghi, **P. Odiase**, et al. Prognostic Utility of Tumor Grade and IDH-Mutation Status for Immunotherapy Response in High Grade Glioma: A Systematic Review and Meta-Analysis. *Journal of Clinical Neuroscience* 152, 2026.
3. W. McKay, A. Gopakumar, S. Bhatia, P. Hartung, D. Lee, A. Pathakamuri, **P. Odiase**, G. A. Zenonos. Prognostic Impact of Multi-Divisional Trigeminal Neuralgia on Pain Outcomes Following Microvascular Decompression. *Clinical Neurology and Neurosurgery*, 2026.
4. S. El Masry, C. A. Ramirez, A. R. Bechtle, **P. Odiase**, E. T. Haupt. Comparative Biomechanical Analysis of Six Orthopedic Fixation Constructs Under Cyclic Loading. *Journal of Orthopaedic Research* 44(2), 2026.
5. S. O'Leary, T. Newsome-Cuby, **P. Odiase**, et al. Spina Bifida in Africa: A Systematic Review and Meta-Analysis of Presentation, Management, and Outcomes. *Journal of Clinical Neuroscience* 148, 2026.
6. S. O'Leary, M. A. Haider, N. Truong, et al., **P. Odiase**, et al. Stereotactic Laser Ablation for Pediatric Central Nervous System Tumors: A Systematic Review and Meta-Analysis. *Journal of Neurosurgery: Pediatrics* 36(2), 2025.
7. A. Ebada, N. Bever, C. J. Carron, **P. Odiase**, C. Wolfe, M. Al Tamimi, S. G. Aoun. Comparing Gabapentin and Pregabalin for Perioperative Pain Management in Lumbar Spine Surgery: A Systematic Review and Meta-Analysis. *World Neurosurgery* 201, 2025.
8. D. B. Kenchappa, O. Korolkova, N. Sakwe, **P. Odiase**, M. G. Izban, A. Sakwe, J. Ochieng. Fetuin-A Modulates Tumor Growth and Invasion in a Basal-Like Triple Negative Breast Cancer Cell Line, MDA-MB-468. *Journal of Pharmacy and Pharmacology Research* 9, 2025.
9. S. O'Leary, N. Fredricks, **P. Odiase**, et al. Foix-Alajouanine Syndrome: A Systematic Review and Meta-Analysis of Presentation, Management, and Outcomes. *Neurochirurgie*, 2025.
10. K. Darko, G. Simmons, W. E. Yevudza Jr, et al., **P. Odiase**, et al. Presentation, Management and Outcomes of Ruptured Intracranial Aneurysms in Africa: A Systematic Review and Meta-Analysis. *Journal of Clinical Neuroscience* 133, 2025.
11. S. O'Leary, W. E. Yevudza Jr, **P. Odiase**, et al. Presentation, Management, and Outcomes of Central Nervous System Metastases in Africa: Systematic Review and Meta-Analysis. *Neuro-Oncology Advances* 7(1), 2025.
12. S. O'Leary, S. Chilukuri, **P. Odiase**, U. Barrie. Global Neurotrauma Registries: Challenges and Opportunities for Low- and Middle-Income Countries. *Journal of Clinical Neuroscience* 131, 2025.
13. E. Kalunta-Crumpton, S. Neerukonda, P. Smith, **P. Odiase**, A. Jenkins, U. Barrie, B. Braga. Dexamethasone Use in Pediatric Neurosurgical Trauma: A Systematic Review and Proportional Meta-Analysis. *Preprint*, 2025.
14. K. Darko, S. Pulido, M. A. Haider, et al., **P. Odiase**, et al. Craniosynostosis in Africa: Insights from Eight Countries — A Systematic Review and Meta-Analysis. *World Neurosurgery*, 2024.

15. K. Darko, I. Shukla, T. Hassan, et al., **P. Odiase**, et al. Presentation, Management, and Outcome of Traumatic Spine Injuries in Africa: A Systematic Review and Meta-Analysis. *Journal of Neurosurgery: Spine* 42(3), 2024.
16. **P. Odiase**, J. Ma, S. Ranganathan, O. Ogunkua, W. B. Turner, D. Marshall, J. Ochieng. The Role of Fetuin-A in Tumor Cell Growth, Prognosis, and Dissemination. *International Journal of Molecular Sciences* 25(23), 2024.
17. **P. Odiase**, H. Terry, A. Banga, K. Rachakonda, A. P. Garg, G. Rachakonda. A Retrospective Community Perspective: Bridging the Health Disparity Gap with Obesity as the Leading Comorbidity in Disparities in Health Outcomes in the COVID-19 Pandemic. *Research Square (preprint)*, 2023.
18. A. R. Banga, K. R. Sekhar, K. J. Rayford, A. Arun, **P. Odiase**, A. P. Garg, M. F. Lima, P. N. Nde, F. Villalta, G. Rachakonda. Xanthine Analogs Suppress *Trypanosoma cruzi* Infection In Vitro Using PDEs as Targets. *Microbiology Research*, 2022.
19. A. R. Banga, **P. Odiase**, K. Rachakonda, A. P. Garg, S. E. Adunyah, G. Rachakonda. Application of C-Terminal *Clostridium perfringens* Enterotoxin in Treatment of Brain Metastasis from Breast Cancer. *Cancers* 14(17), 2022.
20. A. Worley*, **P. Odiase***, M. Cook, A. Britt, G. Rachakonda. COVID-19 Incidence and Mortality in Long-Term Care Facilities in Tennessee. *International Journal of Nursing and Health Care Research*, 2022. (*equal contribution)
21. **P. Odiase**, K. Rachakonda, M. Kress, F. Villalta, G. Rachakonda. Respiratory and Neurological Disease across Different Ethnic Groups Is Influenced by the Microbiome. *Microorganisms* 9(9), 2021.
22. J. A. Hardaway, S. M. Sturgeon, C. L. Snarrenberg, Z. Li, X. Z. S. Xu, D. P. Bermingham, **P. Odiase**, W. C. Spencer, D. M. Miller III, L. Carvelli. Glial Expression of the *Caenorhabditis elegans* Gene *swip-10* Supports Glutamate Dependent Control of Extrasynaptic Dopamine Signaling. *The Journal of Neuroscience* 35(25), 2015.

PUBLISHED ABSTRACTS

1. K. Darko, I. Shukla, T. Hassan, et al., **P. Odiase**, et al. Presentation, Management and Outcome of Traumatic Spine Injuries in Africa: A Systematic Review and Meta-Analysis (Abstract 1117). *Neurosurgery* 71(Suppl 1), 2025.
2. J. Goodwin, T. Rana, **P. Odiase**. Modulation of Plasma Membrane Cholesterol Affects DAT Function in the Presence of Methamphetamine. *Journal of Biological Chemistry* 299(3), 2023.

PRESENTATIONS

1. **P. Odiase**, E. Wu, J. Yap, R. Ghanta, M. Kim, R. Deshpande, S. Kapoor. Machine Learning Prediction of Refeeding Syndrome in Critically Ill Patients Using Pre-Feeding Data. Critical care meeting, 2027. Oral Presentation (first author).
2. S. O'Leary, S. Srinivasan, **P. Odiase**, et al. Oncolytic Viruses for Recurrent High Grade Glioma: A Systematic Review and Meta-Analysis. Association of American Neurological Surgeons (AANS). 2025. Boston, MA. Oral Presentation.
3. S. Srinivasan, Y. Ilyas, **P. Odiase**, et al. Low Prevalence of Radionecrosis in Radio-Immunotherapy for Primary Brain Tumors: A Meta-Analysis of Toxicity. AANS. 2025. Boston, MA. Poster Presentation.
4. S. Srinivasan, Y. Ilyas, **P. Odiase**, et al. Are Tumor Vaccines for High Grade Primary Brain Tumors Safe? A Comparative Meta-Analysis. AANS. 2025. Boston, MA. Poster Presentation.
5. S. Srinivasan, **P. Odiase**, et al. Prognostic Utility of Histology and IDH-Mutation Status for Immunotherapy Response in High Grade Glioma: A Hazard Meta-Analysis. AANS. 2025. Boston, MA. Poster Presentation.
6. S. Srinivasan, M. Guirguis, **P. Odiase**, et al. The Impact of Antimicrobial Prophylaxis on the Rate of Ventriculitis in Patients with External Ventricular Drains: A Systematic Review and Network Meta-Analysis. AANS. 2025. Boston, MA. Poster Presentation.
7. O. Ajibola, T. Poulson, **P. Odiase**, et al. The Impact of the COVID-19 Pandemic on 2020 Spine Surgery Caseloads: A Global Systematic Review. AANS. 2025. Boston, MA. Poster Presentation.
8. O. Ajibola, T. Poulson, **P. Odiase**, et al. Comparing Surgical Outcomes in Spine Surgery: A Systematic Review of Pre-Pandemic and Pandemic (2020) Years. AANS. 2025. Boston, MA. Poster Presentation.
9. M. Bah, L. Simo, P. Patel, S. Chilukuri, **P. Odiase**, et al. A Comparison of Liposomal Bupivacaine to Traditional Pain Control for Postoperative Analgesia following Spine Surgery: A Systematic Review and Meta-Analysis. AANS. 2025. Boston, MA. Poster Presentation.
10. M. A. Haider*, C. Kim*, **P. Odiase**, et al. Comparing 3D Exoscope and Operative Microscope Use in Neuro-Oncologic Surgery: An Evaluation of Outcomes and Neurosurgeon Perception Ratings. AANS. 2025. Boston, MA. Poster Presentation.
11. Impact of Concurrent Spine Surgery on Multilevel Spinal Fusion in the Management of Adult Degenerative Scoliosis. AANS. 2025. Boston, MA. Poster Presentation.
12. The Correlation Between Cholesterol and Substance Use Disorders. Research Centers in Minority Institutions (RCMI) Consortium National Conference. 2025.
13. **P. Odiase**, B. Gbenga, O. Kadiri, S. Ajudua, S. Onabanjo, M. Kukoyi. A Cross-sectional Evaluation of a Community Based Intervention for Cardiopulmonary and Accident Resuscitation Education (C.A.R.E.) in Lagos, Nigeria: Opportunities, Challenges and Limitations. African Conference on Emergency Medicine. 2024. Botswana. Oral Presentation.
14. K. Darko, I. Shukla, T. Hassan, et al., **P. Odiase**, et al. Presentation, Management and Outcome of Traumatic Spine Injuries in Africa: A Systematic Review and Meta-Analysis. Congress of Neurological Surgeons (CNS) Annual Scientific Meeting. 2024. Houston, TX. Oral Presentation.

15. **P. Odiase**, J. Ma, R. S. Lavadi, S. Ranganathan, K. Hamilton, N. Agarwal. A Multi-omics Study of Lumbar Disc Herniations. CNS. 2024. Houston, TX. Digital Poster.
16. K. Darko, S. Pulido, M. A. Haider, et al., **P. Odiase**, et al. Craniosynostosis in Africa: A Systematic Review and Meta-Analysis. CNS. 2024. Houston, TX. Digital Poster.
17. K. Darko, G. Simmons, E. Yevudza, et al., **P. Odiase**, et al. Management and Outcome of Ruptured Intracranial Aneurysms in Africa: A Systematic Review and Meta-Analysis. CNS. 2024. Houston, TX. Printed Poster.
18. S. O'Leary, E. Yevudza, M. A. Haider, et al., **P. Odiase**, et al. Brain and Spinal Cord Metastases in Africa: Hospitalizations, Management and Outcomes: Systematic Review and Meta-analysis. CNS. 2024. Houston, TX. Digital Poster.
19. I. Shukla, J. Wang, M. Guirguis, **P. Odiase**, et al. Radiotherapy and Radiosurgery in Africa: A Systematic Literature Review and Proposed Solutions for Resource-Limited Environments. CNS. 2024. Houston, TX. Digital Poster.
20. A. Jenkins, F. Azam, P. Venkatesh, **P. Odiase**, et al. Assessing the Utility of 3D Printing for Implant Development in Spinal Surgery. CNS. 2024. Houston, TX. Digital Poster.
21. A. Jenkins, F. Azam, P. Venkatesh, **P. Odiase**, et al. Evaluating the Role of Intraoperative 3D Printed Guides in Spinal Surgery Outcomes. CNS. 2024. Houston, TX. Digital Poster.
22. J. Wang, P. Venkatesh, R. Hanoon, **P. Odiase**, et al. Spinal Cord Injury Following Surgical Treatment of Adolescent Idiopathic Scoliosis: A Systematic Review of Presentation, Management, and Outcomes. CNS. 2024. Houston, TX. Digital Poster.
23. **P. Odiase**, et al. Evaluating a Community-Based Intervention for Cardiopulmonary and Accident Resuscitation Education in Lagos, Nigeria. Houston Global Health Conference. 2024. Poster.
24. **P. Odiase**, et al. Alterations in Dopamine (DA) Signaling in Mutagenic Variations of the DA Transporter. Association of Nigerian Physicians in the Americas Scientific Conference. 2024. Anaheim, CA. Poster Presentation.
25. **P. Odiase**, et al. The Effect of Androgen-Regulated Oncomirs on PPARγ Levels in Prostate Cancer Cells. Association of Nigerian Physicians in the Americas Scientific Conference. 2024. Anaheim, CA. Poster Presentation.
26. J. Goodwin, T. Rana, **P. Odiase**. Modulation of Plasma Membrane Cholesterol Affects DAT Function in the Presence of Methamphetamine. American Society for Biochemistry and Molecular Biology. 2023. Poster Presentation.
27. G. Rachakonda, **P. Odiase**. Health Disparities in Neurological Outcomes in Patients Infected with COVID-19. Research Centers in Minority Institutions Consortium National Conference. 2023. Poster.
28. **P. Odiase**, et al. Respiratory and Neurological Disease across Different Ethnic Groups Is Influenced by the Microbiome. Association of Nigerian Physicians in Americas Scientific Conference. 2022. Durham, NC. Oral Presentation.
29. Xanthine Analogs Suppress *Trypanosoma cruzi* Infection In Vitro Using PDEs as Targets. Association of Nigerian Physicians in Americas Scientific Conference. 2022. New York, NY. Oral Presentation.
30. **P. Odiase**, et al. Role of *swip-10*, a Novel Gene, in the Regulation of Dopamine Signaling in *C. elegans*. National Conference on Undergraduate Research. 2015.

SKILLS

- Python, R Studio, SQL
- JavaScript, HTML5/CSS, Unity
- Artificial Intelligence
- Machine Learning & Artificial Neural Networks
- Image Detection Programming
- Mathematical Modeling & Statistical Analysis
- EEG, Facial-Affect Tracking, Pupillometry
- Excel/Power BI, VBA
- STORM Microscopy & DBSCAN
- Optogenetics, SWIP Assays
- *C. elegans* Husbandry, Cell Culture

HONORS & AWARDS

- Honorable Mention Award (top 5%), ACM CHI 2026
- Valedictorian, Fisk University
- Erastus Milo Cravath Presidential Scholarship
- Phi Beta Kappa Honor Society
- Sigma Xi – The Scientific Research Honor Society
- Beta Kappa Chi Scientific Honor Society
- Gold Key Chapter of Mortar Board National College Senior Honor Society
- Alpha Mu Gamma Foreign Language Honor Society

CLINICAL SHADOWING

- University of Pittsburgh Medical Center
- Tristar Medical Center, Neurosurgery
- Truman Medical Center, Bluford Healthcare Leadership Institute, Neurosurgery
- Nashville General Hospital, Ophthalmology and General Surgery
- Salt Wagon Clinic, Meharry Medical College, Internal Medicine